

CURRICULUM VITAE

Rommel Real

Associate Professor

Department of Mathematics, Physics, and Computer Science

College of Science and Mathematics

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Education

August 2016 - July 2021	PhD Science, Mathematical Sciences Institute, <i>The Australian National University</i> Title of PhD dissertation: Convergence Results for Variational Regularization and Landweber Iteration Under Heuristic Rules Supervisor: Dr. Qinian Jin
November 2012 - June 2015	Master of Science in Applied Mathematics (major in Optimization and Approximation), Institute of Mathematics, <i>University of the Philippines Diliman</i> Title of Masters Thesis: Moreau Envelope-based Method for Compressed Sensing Supervisor: Dr. Fredegusto Guido David
June 2005 - 2009	Bachelor of Science in Applied Mathematics (major in Operations Research), <i>University of the Philippines Mindanao</i>

Work Experience

August 2015 - present	Associate Professor Department of Mathematics, Physics, and Computer Science, <i>University of the Philippines Mindanao</i> Teaching courses on numerical analysis, optimization, and advanced calculus
January 2015 - July 2015	Lecturer, Institute of Mathematics, <i>University of the Philippines Diliman</i> Handled courses on algebra and trigonometry

Selected Publications

Real, R.R. (2025). A Heuristic Rule for Landweber Iteration in Banach Spaces. In: Chatzakou, M., Ruzhansky, M., Van Bockstal, K. (eds) Direct and Inverse Problems with Applications. DIPARTCSSMA 2024. Trends in Mathematics(), vol 12. Birkhäuser, Cham. https://doi.org/10.1007/978-3-031-98645-1_19

Real, R.R. Hanke-Raus rule for Landweber iteration in Banach spaces. Numer. Math. 156, 345-373 (2024). https://doi.org/10.1007/s00211-023-01389-1
R. Real and Q. Jin. 2020. A revisit on Landweber iteration. Inverse Problems 36 075011 DOI: https://doi.org/10.1088/1361-6420/ab8bc4
H. Liu, R. Real, X. Lu, X. Jia, and Q. Jin. 2020. Heuristic discrepancy principle for variational regularization of inverse problems. Inverse Problems. 36 075013 DOI: https://doi.org/10.1088/1361-6420/ab844a
R. Real. Solving Inverse Problems Interestingly. 2020. International Journal of Applied Sciences and Smart Technologies. 2 (1) DOI: https://doi.org/10.24071/ijasst.v2i1.1972

Research visits

5 June - 14 July 2025	CIMPA-ICTP Research in Pairs fellowship Julius-Maximilians-Universität Würzburg, Germany Supervisor: Prof. Dr. Frank Werner
9 - 26 July 2024	Ghent University, Belgium Host: Dr. Karel van Bockstal

Training, Workshop, and Presentations

3-11 December 2025	Southeast Asian Mathematical Society (SEAMS) School 2025 on Inverse Problems and Optimal Control, University of the Philippines Baguio (co-organizer, member of scientific committee)
1-5 September 2025	Annual Meeting 2025 of the Austrian Mathematical Society and German Mathematical Society, Johannes Kepler University, Linz, Austria Global South Scholarship awardee and minisymposium presenter
22 November - 1 December 2023	Southeast Asian Mathematical Society (SEAMS) School 2023 Scientific Computing for Differential Equations and Applications, University of the Philippines Diliman (Resource speaker)
1-5 August 2022	The East and Southeast Asia Workshop on Inverse Problems and Optimal Control (online)
15 May 2021	Annual Convention of the Mathematical Society of the Philippines Oral presentation: <i>Heuristic discrepancy principle for Landweber iteration in Banach spaces</i>
1-12 July 2019	Applied Inverse Problems Conference Université Grenoble-Alpes, Grenoble, France Contributed talk: <i>Hanke-Raus Rule for Generalized Variational Regularization</i>

12-14 October 2018	International Conference on Inverse Problems Fudan University, Shanghai, China Contributed talk: <i>Hanke-Raus Heuristic Rule for Landweber Iteration with General Nonsmooth Convex Penalty Functional</i>
6-17 August 2018	The 9th International Conference on Inverse Problems and Related Topics National University of Singapore, Singapore Contributed talk: <i>Hanke-Raus Heuristic Rule for Landweber Iteration with General Nonsmooth Convex Penalty Functional</i>
26 June - 7 July 2017	Australian Mathematical Sciences Institute (AMSI) Winter School on Computational Foundations of Data Science Queensland University of Technology, Brisbane City, Australia
11-13 April 2017	Science Communication Workshop Australian National Centre for the Public Awareness of Science, The Australian National University
9 January - 3 February 2017	AMSI Summer School in the Mathematical Sciences University of Sydney, New South Wales, Australia
7-15 September 2015	Southeast Asian Mathematical Society (SEAMS) School: Modelling and Simulation for the Environmental Phenomena Universitas Sanata Dharma, Yogyakarta, Indonesia

Work to Popularize Research

2025	An Introduction to Inverse Problems CIMPA-ICTP online course recorded las July 2025 at CIRM - Centre International de Rencontres Mathématiques. Info: https://www.carmin.tv/en/collections/an-introduction-to-inverse-problems
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Skills and Competencies

	Experienced with using Matlab and Python, Competent with LaTeX and most Microsoft Office programs
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